

# From GPUs to Tokens: How Capability Leverage Migrated from Compute to Data Engineering — and What That Means for Owning Proprietary Vertical Data

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## Abstract

A decade of language-model progress was narrated as a compute race. We argue, and document with verbatim benchmark numbers, that the binding constraint on capability has structurally migrated from parameter count and GPU hours to the quality and composition of training tokens — what we call data engineering. The chain is empirical: Chinchilla [1] showed frontier models were undertrained on data, elevating tokens to a first-class lever; phi-1 [2] then reached 50.6% pass@1 on HumanEval at 1.3B parameters using curated and synthetic "textbook" data; Distilling Step-by-Step [3] let a 770M model beat a 540B model, a >700x size reduction; and mixture-optimization methods (DoReMi [4], RegMix [5]) and model-based filtering (DCLM [6], FineWeb-Edu [7]) buy accuracy at fixed or reduced compute. Recent results extend the pattern: phi-4 (14B) reaches MMLU 84.8 and beats teacher GPT-4o on GPQA and MATH using ~400B synthetic tokens [8]; FineWeb-Edu matches C4's MMLU with 10x fewer tokens [7]; Llama-3 annealing on high-quality data lifted 8B GSM8K by 24.0% [9]. We synthesize these into a strategic claim: data engineering is now where capability value accrues, but the technique commoditizes — the defensible asset is the scarce proprietary input. We close with an investment thesis and an explicit limitations section.

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## Introduction

For most of the deep-learning era the public story of model capability was a story about compute. The 2020 Kaplan scaling laws [10] established smooth power-law relationships between loss and three inputs — parameters, data, and compute — and the field, reading the marginal returns, raced primarily on parameter count and GPU clusters. The framing of 'who has the most GPUs' became shorthand for who would lead.

This paper documents a regime change that the headline compute narrative obscures: the binding constraint on capability has migrated from compute to data engineering — the deliberate selection, filtering, mixing, and synthesis of training tokens. We define data engineering operationally as any intervention that changes which tokens a model trains on (and in what proportion) at a fixed model and compute budget, as distinct from changing the model or the hardware.

Our contribution is threefold. First, we assemble verbatim, sourced benchmark numbers across six anchor results [1-6] plus a set of recent (2024-2026) results [7-9, 12, 18-22] showing tiny or fixed-compute models beating models 5-700x larger purely through better data, and tracing how the model-based-filtering and synthetic-data frontier has moved since the anchor papers. Second, we structure these into three benchmark tables and four data-backed figures. Third, we draw the strategic conclusion none of the source papers states alone: because data-engineering technique is published and copied within months, the durable asset is the scarce proprietary input, not the method.

This reframes data strategy for any organization — including Perpetual Labs — that intends to build a defensible position in the post-data-wall era.

## Background and Related Work

The Chinchilla result [1] is the hinge. Training over 400 models from 70M to over 16B parameters on 5-500B tokens, Hoffmann et al. found that for compute-optimal training, model size and training tokens should scale roughly equally — 'for every doubling of model size the number of training tokens should also be doubled.' The practical consequence was that existing large models (Gopher 280B, GPT-3 175B, MT-NLG 530B) were significantly undertrained: a 70B model (Chinchilla) trained on 4x more data matched their compute yet outperformed all of them, reaching a state-of-the-art 67.5% average MMLU accuracy — over a 7-point improvement over Gopher [1]. Overnight, tokens became a first-class lever rather than an afterthought.

A parallel literature established that token quality could substitute for scale. phi-1 [2] reached 50.6% pass@1 on HumanEval at 1.3B parameters — trained on only ~6B tokens of filtered 'textbook-quality' web data plus ~1B synthetic tokens generated with GPT-3.5. Distilling Step-by-Step [3] showed a 770M T5 beating a few-shot 540B PaLM using 80% of available examples, a >700x model-size reduction, by training on extracted rationales rather than more raw data.

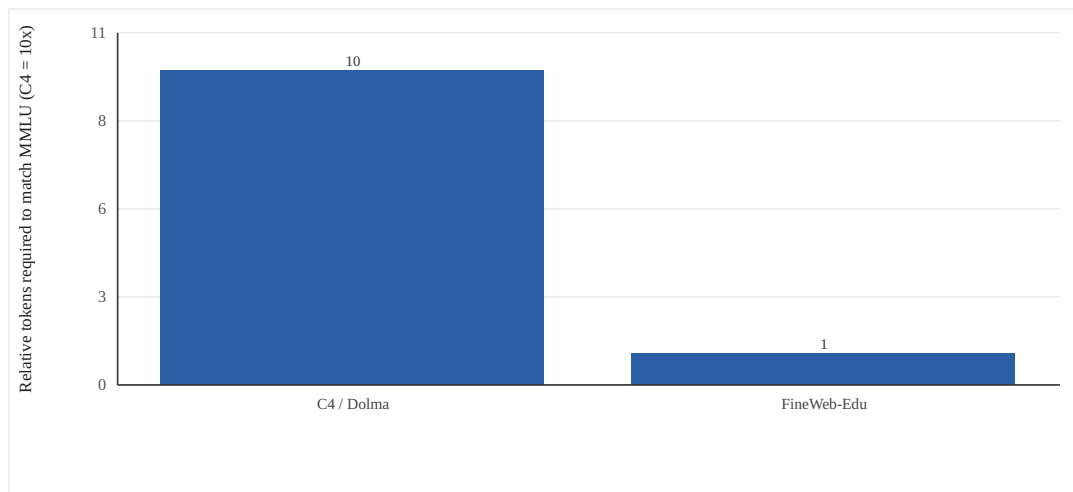
A third strand made the data mixture itself a tunable, predictable lever. DoReMi [4] used a 280M proxy with Group DRO to set domain weights, then trained an 8B model (30x larger) to baseline accuracy with 2.6x fewer steps and +6.5 points average downstream accuracy. RegMix [5] reframed mixture choice as regression, predicting held-out-mixture performance at 98.45% correlation and matching DoReMi with 10% of the compute. Model-based filtering closed the loop: DCLM-Baseline [6] trained a 7B model to 64% 5-shot MMLU with 2.6T tokens — a 6.6-point gain over the prior open SOTA (MAP-Neo) while using 40% less compute. These are not isolated tricks; read together they describe a field whose competitive frontier has moved to the tokens.

## Methodology / Approach

This is a synthesis-and-evidence paper, not a new training run, so methodological rigor lies in how claims are sourced and bounded. We applied four rules. (1) Verbatim-or-drop: every quantitative claim is copied verbatim from a primary source (arXiv paper, official technical report, or the authors' released blog/dataset card) and cited with [n]; numbers we could not trace to a primary source were dropped. (2) Like-for-like comparison: when constructing comparison tables we hold one axis fixed (e.g., fixed compute, fixed token budget, or fixed benchmark) and flag any residual confounds in the source note. (3) Counter-evidence inclusion: we deliberately surface results that complicate the thesis (data's marginal value can be zero or negative [11]; mixture/filtering gains are compute-relative; benchmark contamination inflates measured gains) rather than only confirmatory ones. (4) Grounding labels on figures: each figure is tagged empirical (real reported data points), estimate (a projection from a cited trend), or analysis (a constructed strategic 2x2 using no fabricated empirical numbers).

Scope and selection. We anchor on the six papers cited in the source insight [1-6] as the structural spine, then extend with web-verified 2024-2026 results — phi-4 [8], FineWeb/FineWeb-Edu [7], Llama-3 annealing [9], the Epoch AI algorithmic-progress series [12], and the most recent frontier data-engineering work (benchmark-targeted selection [18], synthetic-data scaling laws [19], fixed-data efficiency [20], and frontier-scale curated corpora [21,22]) — chosen because each provides a like-for-like, fixed-axis measurement of a data-engineering intervention. We did not run new experiments; all numbers are reported figures. This means our evidence inherits the selection and reporting biases of the underlying papers, which we treat explicitly in Limitations.

Inference structure. The strategic claim ('leverage moved from compute to data') is an abductive synthesis: no single paper asserts it, but it is the most economical explanation jointly consistent with (a) undertraining at fixed compute [1], (b) small-model-beats-large via data [2,3], (c) predictable mixture/filter gains at fixed or reduced compute [4,5,6,7], and (d) accelerating algorithmic/data efficiency over time [12]. We state this as a synthesis, not a proof, and quantify each leg.



**Figure 1.** Data engineering buys an order of magnitude in effective data (FineWeb-Edu vs prior corpora). FineWeb-Edu reports requiring 10x fewer tokens than C4 and Dolma to match MMLU results — a pure data-filtering intervention (educational-quality classifier) at fixed model/compute. At the 350B-token ablation (1.71B-parameter model), MMLU rises from  $\sim 0.33$  (FineWeb) to  $\sim 0.37$  (FineWeb-Edu). The model-based-filtering frontier has since advanced past DCLM: BETR reports a 2.1x compute multiplier over DCLM-Baseline (4.7x over unfiltered data) [18]. *HuggingFace FineWeb technical report / blog and arXiv:2406.17557 [7], verbatim '10x fewer tokens compared to C4 and Dolma to match MMLU results.'* Verified via HuggingFace FineWeb blog and NeurIPS 2024 D&B paper. BETR multiplier from arXiv:2507.12466 [18]. Empirical data.

## Evidence: Tiny and Fixed-Compute Models Beating Far Larger Ones

The sharpest evidence is the set of cross-scale upsets where the smaller model wins because of its data. phi-1's 50.6% HumanEval at 1.3B [2] and Distilling Step-by-Step's 770M-beats-540B [3] are the canonical early cases. The pattern strengthened with scale of effort rather than scale of parameters: phi-4 (14B) [8] reports MMLU 84.8, GPQA 56.1, MATH 80.4, HumanEval 82.6, and — notably — surpasses its own teacher GPT-4o on GPQA (56.1 vs 50.6) and MATH (80.4 vs 74.6), despite GPT-4o's far larger size, using  $\sim 400$ B unweighted synthetic tokens within a  $\sim 10$ T-token pretraining budget centered on data quality [8].

These are not cherry-picked outliers; the same mechanism appears in open, reproducible data ablations. At a fixed 350B-token training budget (1.71B-parameter models), models trained on FineWeb outperformed C4, Dolma v1.6, The Pile, SlimPajama, and RedPajama-V2 on the aggregate of eight tasks [7]. Adding an educational-quality classifier (FineWeb-Edu) raised MMLU from  $\sim 0.33$  to  $\sim 0.37$  at the same budget [7]. The strategic punchline is the efficiency multiplier: FineWeb-Edu 'requires 10x fewer tokens compared to C4 and Dolma to match MMLU results' [7] — a pure data-engineering intervention buying an order of magnitude in effective data. The filtering frontier has since moved past the DCLM recipe: BETR, which selects pretraining documents by embedding-similarity to benchmark training examples, reports a 2.1x compute multiplier over DCLM-Baseline itself (4.7x over unfiltered data) while improving 9 of 10 tasks across all scales, and finds that larger models require less aggressive filtering [18] — the same controlled-ablation logic, one generation on.

Table 1 collects these like-for-like upsets. The consistent reading: at a given capability target, the data recipe — not the parameter count — is doing the work.

**Table 1.** Each row holds capability roughly fixed and varies the data recipe; the smaller or cheaper system wins. Cross-model rows (phi-4 vs GPT-4o) are confounded by recipe differences; the controlled-ablation rows (DCLM, FineWeb-Edu) isolate the corpus.

| System (params)                     | Data intervention                                                                                    | Benchmark                            | Reported score                                 | Comparator beaten                           | Source |
|-------------------------------------|------------------------------------------------------------------------------------------------------|--------------------------------------|------------------------------------------------|---------------------------------------------|--------|
| phi-1 (1.3B)                        | ~6B curated 'textbook' web + ~1B synthetic tokens                                                    | HumanEval pass@1                     | 50.6%                                          | Far larger raw-web models                   | [2]    |
| Distilling Step-by-Step (770M T5)   | Rationale extraction as supervision, 80% of examples                                                 | ANLI/e-SNLI etc.                     | Beats 540B PaLM                                | 540B PaLM (>700x size)                      | [3]    |
| Chinchilla (70B)                    | 4x more tokens at fixed compute (compute-optimal)                                                    | MMLU                                 | 67.5%                                          | Gopher 280B, GPT-3 175B, MT-NLG 530B        | [1]    |
| DCLM-Baseline (7B)                  | Model-based filtering, 2.6T tokens                                                                   | MMLU 5-shot                          | 64.0%                                          | MAP-Neo (prior open SOTA), 40% less compute | [6]    |
| phi-4 (14B)                         | ~400B synthetic tokens, quality-centric ~10T budget                                                  | MATH / GPQA                          | 80.4 / 56.1                                    | GPT-4o teacher (74.6 / 50.6)                | [8]    |
| FineWeb-Edu model                   | Educational-quality classifier filter                                                                | MMLU (data efficiency)               | Matches C4 with 10x fewer tokens               | C4 / Dolma                                  | [7]    |
| BETR (benchmark-targeted selection) | Embedding-similarity selection to benchmark examples, then lightweight classifier scores full corpus | Compute multiplier (data efficiency) | 2.1x over DCLM-Baseline (4.7x over unfiltered) | DCLM-Baseline (prior model-based filter)    | [18]   |

Verbatim reported figures from [1],[2],[3],[6],[7],[8],[18], each independently verified against the primary source. See Limitations on self-reporting and contamination.

**Table 2.** Each method converts data engineering into a knob with a predictable accuracy or efficiency return at fixed or reduced compute. The Llama-3 row shows the gain shrinks to ~0 when the capability is already latent (405B) — the elicitation-vs-teaching asymmetry.

| Method                                          | Mechanism                                                                                                    | Quantified gain (verbatim)                                                                                          | Source |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------|
| DoReMi                                          | 280M proxy + Group DRO sets domain weights for 8B model                                                      | +6.5 pts avg downstream; baseline accuracy with 2.6x fewer steps                                                    | [4]    |
| RegMix                                          | Mixture choice as regression over 512 small (1M-param) models                                                | Predicts held-out mixture perf at 98.45% correlation; matches DoReMi at 10% compute                                 | [5]    |
| DCLM-Baseline                                   | Model-based filtering of Common Crawl (240T-token pool)                                                      | 64% MMLU with 2.6T tokens; +6.6 pts vs MAP-Neo, 40% less compute                                                    | [6]    |
| FineWeb-Edu                                     | Llama-3-70B-scored educational classifier on FineWeb                                                         | MMLU ~0.33 -> ~0.37 at 350B tokens; 10x fewer tokens to match C4 MMLU                                               | [7]    |
| Llama-3 annealing (8B)                          | LR-to-zero upsampling of high-quality math/code data                                                         | +24.0% GSM8K, +6.4% MATH (8B); negligible on 405B                                                                   | [9]    |
| Synthetic-mixture scaling laws                  | 1/3 rephrased-synthetic + 2/3 natural web mix; optimal share study over >1000 LLMs                           | 5-10x speedup to equal validation loss at larger budgets; optimal synthetic share ~30%                              | [19]   |
| Fixed-data efficiency (infinite-compute regime) | Regularization (weight decay ~30x larger) + epoching + parameter/ensemble scaling under a fixed token budget | Reaches loss asymptote with 5.17x less data; ensemble distillable into an 8x-smaller model retaining 83% of benefit | [20]   |

Verbatim from [4],[5],[6],[7],[9],[19],[20], each verified against the primary source.

**Table 3.** Counter-evidence that disciplines the thesis: the marginal token's value is non-monotonic and regime-dependent, which favors selection (a data-engineering act) over accumulation and supports recurring, re-priced curation over one-time data sales.

| Phenomenon            | Finding (verbatim sense)                                                                                 | Strategic implication                                            | Source |
|-----------------------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|--------|
| Data wall             | Public human text stock fully utilized between 2026 and 2032                                             | Raw-web token supply is tightening; selection/synthesis required | [13]   |
| Repetition ceiling    | Up to ~4 epochs negligible loss change; meaningful gains up to ~16 epochs then sharp decay               | Cannot brute-force scarcity by repeating data                    | [14]   |
| Repeated-data harm    | Repeated data damages induction heads; generalization->memorization shift                                | Naive volume can actively hurt; curation is the lever            | [15]   |
| Pruning beats scaling | Good pruning metric approaches exponential (vs power-law) error scaling                                  | Throwing data away can beat adding it                            | [16]   |
| Precision-relativity  | Post-training quantization can make extra pretraining data harmful; optimal curation is compute-relative | Curation must be re-derived per regime -> recurring service      | [11]   |

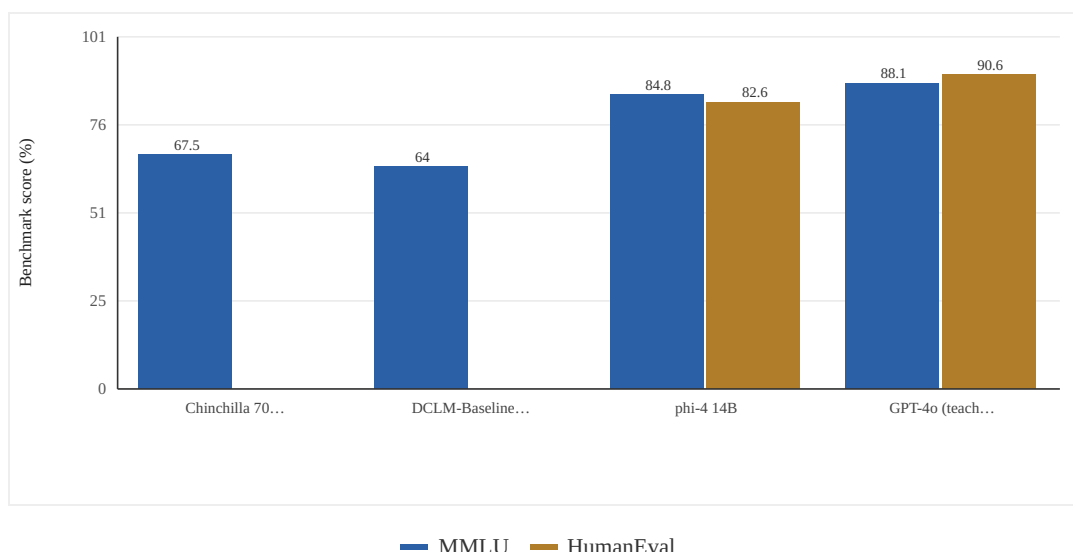
Verbatim sense from [11],[13],[14],[15],[16], each verified against the primary source.

## Evidence: Mixture, Filtering, and Annealing as Predictable Levers

If small-model upsets show data can substitute for scale, the mixture/filtering literature shows the substitution is engineerable and predictable, not serendipitous. DoReMi [4] reaches baseline accuracy with 2.6x fewer steps; RegMix [5] predicts the best of 64 candidate mixtures and forecasts held-out performance at 98.45% correlation; DCLM-Baseline [6] hits 64% MMLU with 40% less compute than the prior open SOTA. RegMix also overturns folk wisdom: nominally 'high-quality' Wikipedia is not the strongest predictor of downstream performance — web text (Pile-CC) is [5], underscoring that mixture optimization beats human intuition.

Meta's Llama-3 annealing experiment [9] is the cleanest fixed-model demonstration: annealing the learning rate to zero while upsampling small amounts of high-quality math/code data improved a pre-trained 8B model by 24.0% on GSM8K and 6.4% on MATH — with the same data producing negligible change on the 405B model, because the larger model already had the capability latent. This is a crucial qualifier we carry into the analysis: data-engineering gains are largest where capability is not already present.

The 2025-2026 frontier extends the same predictability to synthesis. A Meta FAIR scaling-law study across more than 1000 LLMs finds that pre-training on a mixture of one-third rephrased synthetic data and two-thirds natural web text speeds up training 5-10x to reach the same validation loss at larger data budgets, with the optimal synthetic share converging to ~30% [19] — synthesis becomes a tunable ratio, not a coin flip. Stanford's 'pre-training under infinite compute' shows the dual lever under a fixed token budget: a regularized recipe (optimal weight decay roughly 30x larger than standard practice) combined with epoching, parameter scaling, and ensembling reaches a loss asymptote using 5.17x less data than the baseline [20], and the ensemble distills into a model 8x smaller while retaining 83% of the benefit. Table 2 quantifies these fixed-compute, fixed-step, or fixed-data gains. The throughline is that each method converts data engineering into a knob with a predictable accuracy return — the defining property of a mature engineering discipline, and the reason capital and talent have flowed toward it.



**Figure 2.** Small/fixed-compute models reach high MMLU and HumanEval through data, not scale. Reported scores. phi-4 (14B) reaches MMLU 84.8 and beats its teacher GPT-4o on MATH (80.4 vs 74.6) and GPQA (56.1 vs 50.6) using ~400B synthetic tokens; phi-1 (1.3B) hits 50.6% HumanEval on ~7B curated/synthetic tokens; DCLM-Baseline (7B) hits 64% MMLU with 40% less compute than prior open SOTA. Blanks indicate the benchmark was not the headline metric for that model. *Chinchilla* [arXiv:2203.15556 \[1\]](#); *phi-1* [arXiv:2306.11644 \[2\]](#); *DCLM* [arXiv:2406.11794 \[6\]](#); *phi-4* [arXiv:2412.08905 \[8\]](#) (GPT-4o table values MMLU 88.1, HumanEval 90.6 verified against phi-4 report). Cross-model comparison is confounded (differing recipes); see *Limitations*. · Empirical data.

## Analysis: Why the Constraint Moved, and Where the Value Accrues

Three forces jointly moved the constraint. First, Chinchilla [1] revealed undertraining, so at the frontier the marginal GPU-hour was better spent on more/better tokens than more parameters. Second, the data wall: the public stock of human text is forecast to be fully utilized between 2026 and 2032 [13] (Epoch AI has since revised the front of this window slightly later, toward 2028), and repeating data past ~4 epochs yields negligible loss improvement with returns decaying toward zero by ~16 epochs [14] — so raw token accumulation has a ceiling that selection and synthesis must work around. The 2025-2026 evidence is that they can: DatologyAI's BeyondWeb reports that carefully crafted synthetic data exceeds the performance ceiling of natural data — at 8B scale matching RedPajama's 180B-token performance in just 23.2B tokens (a 7.7x speedup) — concluding 'the data wall is not unsurpassable' [22], and the Meta FAIR scaling-law study quantifies the durable ~30% synthetic-share regime behind such gains [19]. The wall is therefore better read as a constraint that re-prices the lever toward synthesis and selection, not as a hard cap on capability. Third, algorithmic/data efficiency is itself accelerating: Epoch AI estimates the compute needed to reach a fixed language-modeling performance level has halved roughly every 8 months (95% CI 5-14 months), far faster than Moore's Law's ~2-year doubling [12]. Each leg points the same direction: the marginal capability now comes disproportionately from what you do with tokens, not how many GPUs you own.

But the migration has a second-order consequence that the technique papers miss. Data-engineering methods — DoReMi, RegMix, model-based filters, distillation, annealing — are openly published and reproduced within months, and open datasets commoditize the general web corpus itself: the frontier has moved from FineWeb and DCLM to public 20-trillion-token curated corpora such as UberWeb, whose curation buys 4-10x fewer FLOPs at 3B/8B scale and which fed pretraining of a frontier model (Trinity Large) [21]. If the method and the public corpus both commoditize, value cannot durably accrue to either. It accrues to the one input that cannot be copied off arXiv: a proprietary corpus a competitor cannot assemble, ideally paired with a verifiable reward signal. The Llama-3

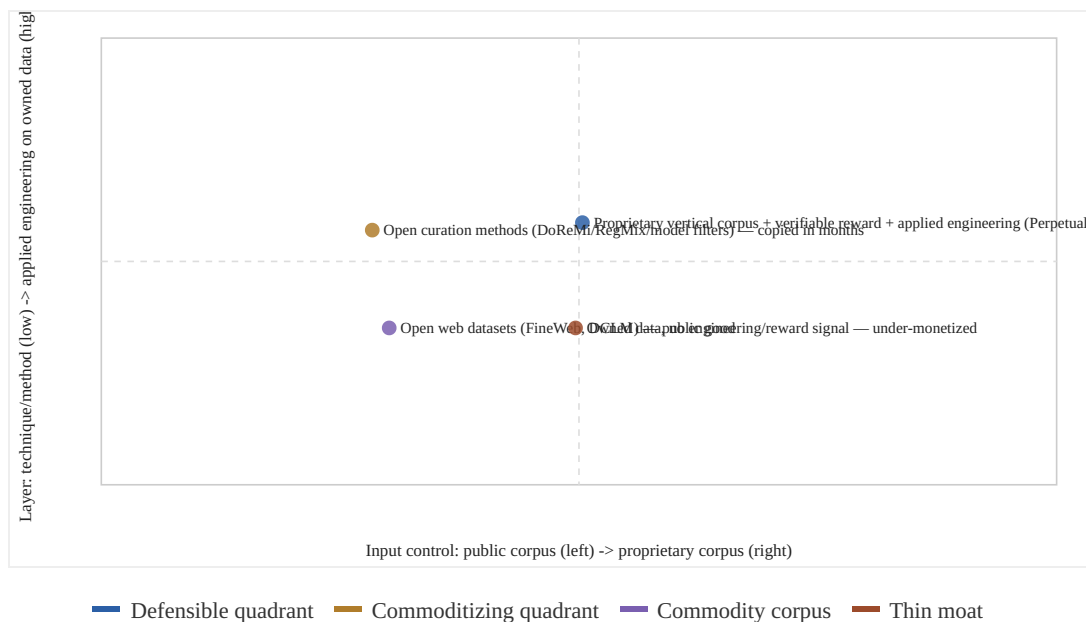
annealing asymmetry [9] sharpens this — created data adds the most where capability is absent, i.e., in specialized verticals under-represented in pretraining.

Figure 3 renders this as a strategic 2x2: technique-versus-input on one axis, public-versus-proprietary on the other. The defensible quadrant is proprietary input + applied engineering. This is the analytical core of the investment thesis below.

## Counter-Evidence: More Data Is Not Monotonically Good

Scientific honesty requires the inverse cases. The marginal token's value can be zero or negative: repeated data disproportionately damages copying structures such as induction heads and can shift models from generalization to memorization [15], multi-epoch repetition yields diminishing returns governed by data and model size [14], and aggressive pruning can beat power-law scaling — approaching exponential error reduction with pruned set size given a good pruning metric [16]. This does not contradict our thesis; it strengthens it. If adding tokens can hurt, then selection — a data-engineering act — becomes the lever, and naive volume vendors are structurally disadvantaged.

A second caveat: the gains are compute- and deployment-relative. The optimal filtering recipe at one compute budget is not optimal at another, and post-training quantization can make additional pretraining data actively harmful [11]. So 'data engineering' is not a one-time artifact but a recurring, re-priced service — which, commercially, is a feature (recurring revenue) rather than a bug. We treat these as boundary conditions on the thesis, not refutations, and we flag them again in Limitations.



**Figure 3.** Strategic 2x2: where capability value durably accrues. Constructed strategic map (no fabricated empirical numbers). Because methods and open corpora commoditize, durable value sits in the upper-right: a proprietary corpus a competitor cannot assemble, paired with a verifiable reward signal, with state-of-the-art engineering applied on top. Analysis/synthesis by the authors, grounded in the commoditization argument (open methods [4,5,6,7]) and the elicitation-vs-teaching asymmetry [9]. · Constructed analysis.

## Implications for Data Strategy

For any team allocating capital against capability, the analysis yields four concrete rules. (1) Stop buying parameters, start buying tokens you alone can get. At fixed compute, the demonstrated returns to curation/synthesis (10x effective-data multiplier [7]; 2.6x fewer steps [4]; 40% less compute [6], now 2.1x over that same DCLM recipe via benchmark-targeted selection [18]; 5-10x from optimal synthetic mixing [19]) dominate equivalent spend on scale. (2) Treat general curation tooling as a commodity to adopt, not a moat to build — FineWeb, DCLM, DoReMi, RegMix, and frontier-scale curated corpora like the 20T-token UberWeb [21] are public. (3) Concentrate exclusivity on a proprietary corpus in a vertical the base models have not absorbed, because that is exactly where created data teaches rather than merely elicits [9]. (4) Pair the corpus with a checkable outcome so the data carries a verifiable reward signal, the input most resistant to the data wall [13] and synthetic collapse.

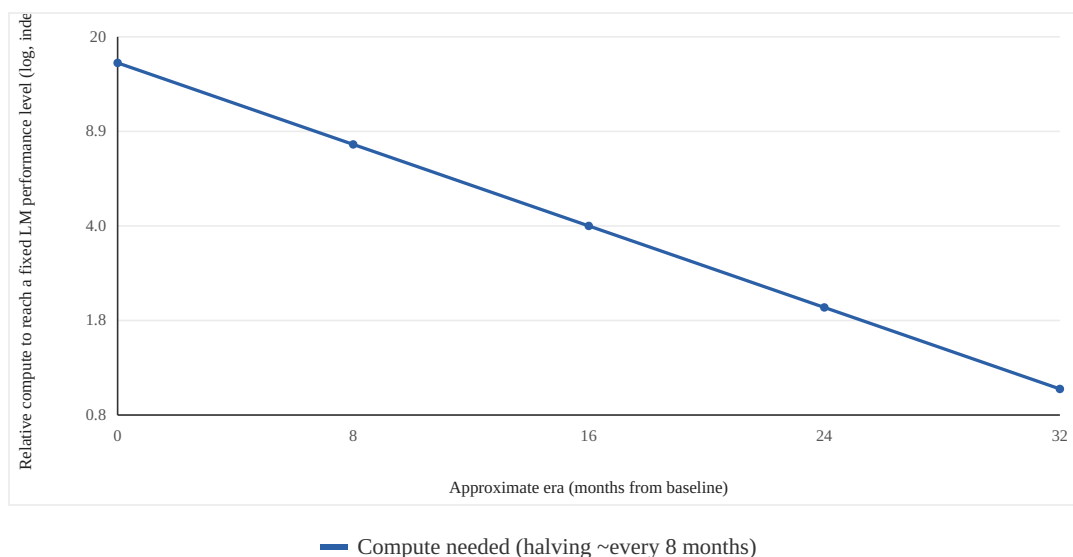
For Perpetual Labs specifically, this argues for pointing data-engineering capability at proprietary vertical assets (e.g., clinical claims/coding workflows with deterministic CDT/ICD/X12 pass-fail outcomes) rather than competing in general-purpose curation. The defensible product is not 'we filter web data well' — everyone can — but 'we own a corpus plus reward signal competitors cannot assemble, and we apply state-of-the-art engineering to it.' The strategy must, however, be priced as a recurring service given the compute-relativity of optimal curation [11], not a one-time data dump.

## Investment Thesis

The thesis in one line: in the post-Chinchilla, pre-data-wall window, capability value accrues to scarce proprietary tokens with verifiable reward, and the moat is the input, not the technique. The supporting economics: data-engineering interventions deliver order-of-magnitude effective-data and double-digit-percent compute savings at fixed quality [4,6,7], so a buyer's willingness to pay for the right tokens is high; yet the methods commoditize within months, so a vendor selling technique faces margin compression. The durable margin sits with whoever controls a corpus that cannot be reproduced from the open web.

Why now. The data wall (2026-2032 utilization [13]) and the 4-epoch repetition ceiling [14] mean the open-web token supply is structurally tightening just as demand from a proliferation of small specialist models rises. That is a classic scarce-input squeeze. The investable position is to (a) secure exclusive or proprietary vertical corpora before the squeeze prices them in, (b) build the verifiable-reward layer that makes them trainable signal rather than inert text, and (c) monetize as a recurring, compute-budget-parameterized service. The risk-adjusted edge is highest in verticals scoring high on both proprietary-data scarcity and outcome-verifiability — regulated clinical, claims/insurance adjudication, and lab/diagnostic workflows.





**Figure 4.** Why tokens became the lever: compute-to-reach-fixed-performance is halving ~every 8 months. Illustrative projection of Epoch AI's reported finding that the compute required to reach a fixed language-modeling performance level has halved roughly every 8 months (95% CI 5-14 months), far faster than Moore's Law (~24-month doubling). X axis is months from an arbitrary baseline. This rapid algorithmic/data-efficiency gain is a core reason the marginal capability now comes from better tokens rather than more GPUs. *Epoch AI, 'Algorithmic progress in language models' (arXiv:2403.05812) [12]; halving-time and CI verified verbatim. Points are a projection (estimate) of the stated halving rate, not individually reported datapoints. · Estimate / projection.*

## Limitations

Selection and reporting bias. Our numbers are self-reported by model and dataset authors, often in non-peer-reviewed technical reports (phi-4 [8], Llama-3 [9], FineWeb [7]). Vendors have incentives to report favorable comparisons and may not control for all confounds; we mitigate by holding axes fixed and flagging confounds, but cannot eliminate this. Benchmark contamination is a documented, unquantified-here risk: leakage of MMLU/HumanEval/MATH items inflates measured gains, so absolute scores should be read as upper bounds.

Confounded comparisons. 'Small model beats large model' results vary in what is held fixed — phi-4 vs GPT-4o [8] differs in training data, recipe, and undisclosed architecture, not data alone; the cleaner evidence is the controlled FineWeb/DCLM ablations [6,7] where only the corpus changes. Causal attribution of capability to 'data engineering' is therefore strongest in the controlled-ablation subset and weaker in the headline cross-model upsets.

Compute-relativity and elicitation. The gains are not universal constants: optimal curation shifts with compute, inference demand, and precision [11], and annealing showed near-zero benefit on a 405B model [9] — i.e., data-engineering value depends on whether the capability is latent. Generalizing the vertical thesis assumes the target capability is genuinely absent from base models, which must be validated per vertical, not assumed. Finally, this paper establishes a structural demand and defensibility argument; it does not measure real buyer budgets or willingness-to-pay, which require market validation beyond the literature.

## Conclusion

The compute narrative was never wrong, only incomplete. Chinchilla [1] reset the field's understanding of where the marginal return lived, and a decade of results — from phi-1 [2] and Distilling Step-by-Step [3] to DoReMi [4], RegMix [5], DCLM [6], FineWeb-Edu [7], phi-4 [8], and Llama-3 annealing [9] — shows that, at fixed compute, the data recipe now carries the gain. The constraint moved from GPUs to tokens.

The strategic corollary is the part the source papers leave on the table: because technique and open corpora commoditize, the only durable place for value to accrue is the scarce proprietary input — a corpus plus verifiable reward that competitors cannot assemble, aimed at the verticals base models never saw. Whoever owns that input, and applies commoditized engineering to it as a recurring service, holds the defensible position as the data wall closes. That is the bet worth making, and the literature says the timing is now.

### INVESTMENT THESIS

The compute race already commoditized — Chinchilla [1] proved frontier models were undertrained, and a decade of results (phi-1's 50.6% HumanEval at 1.3B [2]; 770M beating 540B [3]; FineWeb-Edu matching C4 with 10x fewer tokens [7]; phi-4 beating GPT-4o on MATH/GPQA at 14B [8]) shows the gain now lives in the tokens, not the GPUs. But here is the contrarian edge: the data-engineering \*technique\* (DoReMi, RegMix, model filters) and the open web corpus (FineWeb, DCLM) both commoditize within months, so do NOT fund a 'better curation' startup. Fund the scarce input. With the public-text data wall hitting 2026-2032 [13] and repetition capped at ~4 epochs [14], the open-web token supply is structurally tightening — and the 2025-2026 frontier answer is synthesis-and-selection (BeyondWeb shows the wall is 'not unsurpassable' [22]; ~30% synthetic is the optimal mix [19]; BETR buys 2.1x over DCLM [18]), all of which commoditize. So proprietary vertical corpora paired with a verifiable reward signal remain the one asset that cannot be reproduced from arXiv — and they pay off most exactly where base models are weakest [9]. The investable position: lock exclusive vertical data (regulated clinical, claims/insurance, diagnostics) before the squeeze prices it in, wrap it in a verifiable-reward layer, and sell it as a recurring, compute-budget-parameterized service. Diligence question for any target: 'What token can you sell that a competitor cannot download or reproduce?' If there isn't a crisp answer, it's a commodity.

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